Quantium Task 1 - Data Analysis

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Load libraries

```
# -------------- LOAD LIBRARIES -----------------
library(data.table)
library(readxl)
library(ggplot2)
library(lubridate)
library(readr)
library(stringr)
```

TASK 1: DATA LOADING & CLEANING

##Load data

```
transactionData <- as.data.table(read_excel("QVI_transaction_data.xlsx"))
customerData <- fread("QVI_purchase_behaviour.csv")
```

Clean transaction data

```
transactionData[, DATE := as.Date(DATE, origin = "1899-12-30")]
transactionData <- transactionData[!grepl("salsa", tolower(PROD_NAME))]
transactionData <- transactionData[PROD_QTY != 200]
transactionData[, PACK_SIZE := parse_number(PROD_NAME)]
transactionData[, BRAND := word(PROD_NAME, 1)]
transactionData[BRAND == "RED", BRAND := "RRD"]
transactionData[BRAND == "Natural", BRAND := "NCC"]

# Merge with customer data
names(customerData)[1] <- "LYLTY_CARD_NBR"
data <- merge(transactionData, customerData, by = "LYLTY_CARD_NBR", all.x = TRUE)
```

TOTAL SALES BY SEGMENT

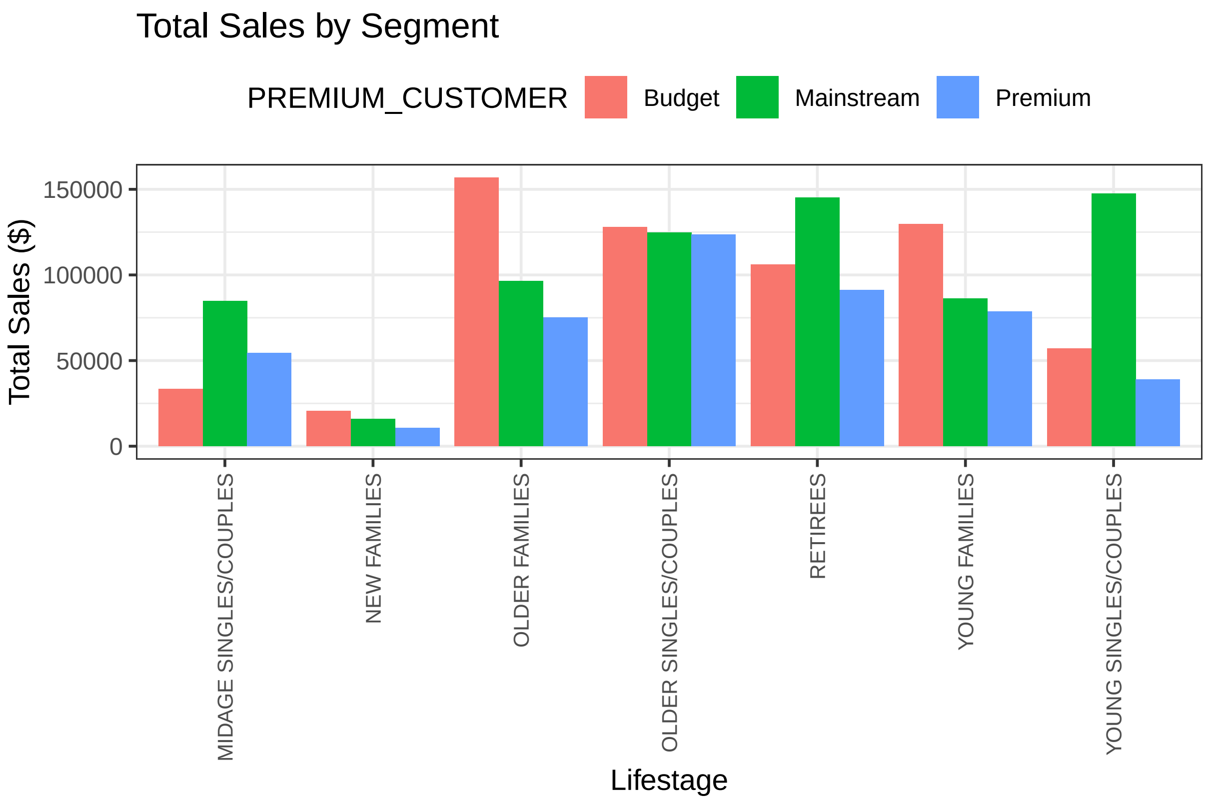
```r
# Aggregate
sales_by_seg <- aggregate(TOT_SALES ~ LIFESTAGE + PREMIUM_CUSTOMER, data = data, FUN = sum)
sales_by_seg <- sales_by_seg[order(-sales_by_seg$TOT_SALES), ]

# Console table
cat("\n===== TOTAL SALES BY SEGMENT =====\n")
```

```
##
## ===== TOTAL SALES BY SEGMENT =====
```

```r
print(sales_by_seg)
```

```
##                 LIFESTAGE PREMIUM_CUSTOMER TOT_SALES
## 3          OLDER FAMILIES           Budget 156863.75
## 14  YOUNG SINGLES/COUPLES       Mainstream 147582.20
## 12               RETIREES       Mainstream 145168.95
## 6          YOUNG FAMILIES           Budget 129717.95
## 4   OLDER SINGLES/COUPLES           Budget 127833.60
## 11  OLDER SINGLES/COUPLES       Mainstream 124648.50
## 18  OLDER SINGLES/COUPLES          Premium 123537.55
## 5                RETIREES           Budget 105916.30
## 10         OLDER FAMILIES       Mainstream  96413.55
## 19               RETIREES          Premium  91296.65
## 13         YOUNG FAMILIES       Mainstream  86338.25
## 8  MIDAGE SINGLES/COUPLES       Mainstream  84734.25
## 20         YOUNG FAMILIES          Premium  78571.70
## 17         OLDER FAMILIES          Premium  75242.60
## 7   YOUNG SINGLES/COUPLES           Budget  57122.10
## 15 MIDAGE SINGLES/COUPLES          Premium  54443.85
## 21  YOUNG SINGLES/COUPLES          Premium  39052.30
## 1  MIDAGE SINGLES/COUPLES           Budget  33345.70
## 2            NEW FAMILIES           Budget  20607.45
## 9            NEW FAMILIES       Mainstream  15979.70
## 16           NEW FAMILIES          Premium  10760.80
```

```r
# Clean chart with 90° rotated labels
p1 <- ggplot(sales_by_seg, aes(x = LIFESTAGE, y = TOT_SALES, fill = PREMIUM_CUSTOMER)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.9)) +
  labs(title = "Total Sales by Segment", y = "Total Sales ($)", x = "Lifestage") +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5, size = 8),
    plot.margin = margin(b = 20, l = 10, r = 10, t = 10),
    legend.position = "top"
  )
print(p1)
```




```r
ggsave("01_total_sales_by_segment.png", p1, width = 12, height = 7, dpi = 300)
```

NUMBER OF CUSTOMERS BY SEGMENT

```r
# Aggregate unique customers
cust_count <- aggregate(LYLTY_CARD_NBR ~ LIFESTAGE + PREMIUM_CUSTOMER,
                        data = data, FUN = function(x) length(unique(x)))
names(cust_count)[3] <- "CUSTOMER_COUNT"
cust_count <- cust_count[order(-cust_count$CUSTOMER_COUNT), ]

# Console table
cat("\n===== NUMBER OF CUSTOMERS BY SEGMENT =====\n")
```

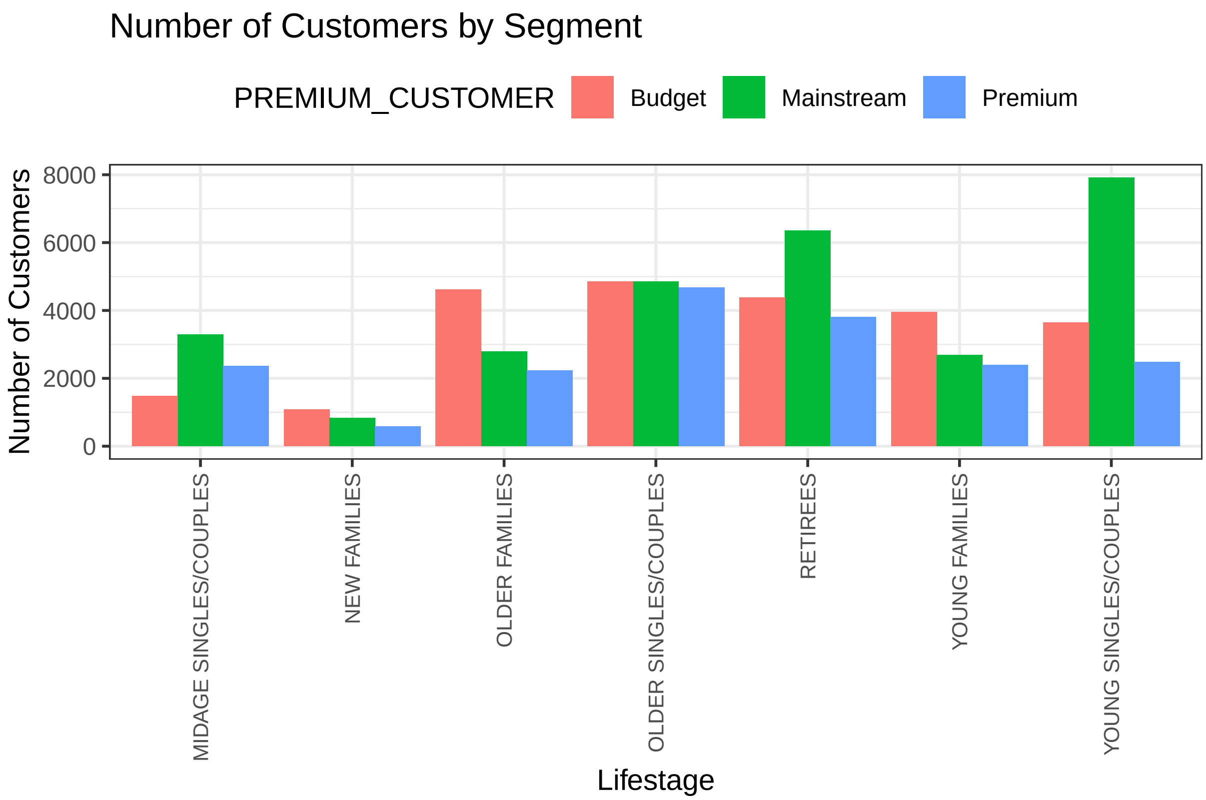
```
##
## ===== NUMBER OF CUSTOMERS BY SEGMENT =====
```

```r
print(cust_count)
```

```
##                  LIFESTAGE PREMIUM_CUSTOMER CUSTOMER_COUNT
## 14  YOUNG SINGLES/COUPLES       Mainstream           7917
## 12                RETIREES       Mainstream           6358
## 11  OLDER SINGLES/COUPLES       Mainstream           4858
```

```
## 4   OLDER SINGLES/COUPLES           Budget          4849
## 18  OLDER SINGLES/COUPLES          Premium          4682
## 3          OLDER FAMILIES           Budget          4611
## 5                RETIREES           Budget          4385
## 6          YOUNG FAMILIES           Budget          3953
## 19               RETIREES          Premium          3812
## 7   YOUNG SINGLES/COUPLES           Budget          3647
## 8  MIDAGE SINGLES/COUPLES       Mainstream          3298
## 10         OLDER FAMILIES       Mainstream          2788
## 13         YOUNG FAMILIES       Mainstream          2685
## 21  YOUNG SINGLES/COUPLES          Premium          2480
## 20         YOUNG FAMILIES          Premium          2398
## 15 MIDAGE SINGLES/COUPLES          Premium          2369
## 17         OLDER FAMILIES          Premium          2231
## 1  MIDAGE SINGLES/COUPLES           Budget          1474
## 2            NEW FAMILIES           Budget          1087
## 9            NEW FAMILIES       Mainstream           830
## 16           NEW FAMILIES          Premium           575
```

```
# Clean chart
p2 <- ggplot(cust_count, aes(x = LIFESTAGE, y = CUSTOMER_COUNT, fill = PREMIUM_CUSTOMER)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.9)) +
  labs(title = "Number of Customers by Segment", y = "Number of Customers", x = "Lifestage") +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5, size = 8),
    plot.margin = margin(b = 20, l = 10, r = 10, t = 10),
    legend.position = "top"
  )
print(p2)
```




```r
ggsave("02_customer_count_by_segment.png", p2, width = 12, height = 7, dpi = 300)
```

AVERAGE PRICE PER UNIT BY SEGMENT

```r
# Calculate unit price
data[, UNIT_PRICE := TOT_SALES / PROD_QTY]

# Aggregate average price
avg_price <- aggregate(UNIT_PRICE ~ LIFESTAGE + PREMIUM_CUSTOMER, data = data, FUN = mean)
avg_price <- avg_price[order(-avg_price$UNIT_PRICE), ]

# Console table (round only numeric column)
avg_price_print <- avg_price
avg_price_print$UNIT_PRICE <- round(avg_price_print$UNIT_PRICE, 2)
cat("\n===== AVERAGE PRICE PER UNIT BY SEGMENT =====\n")
```

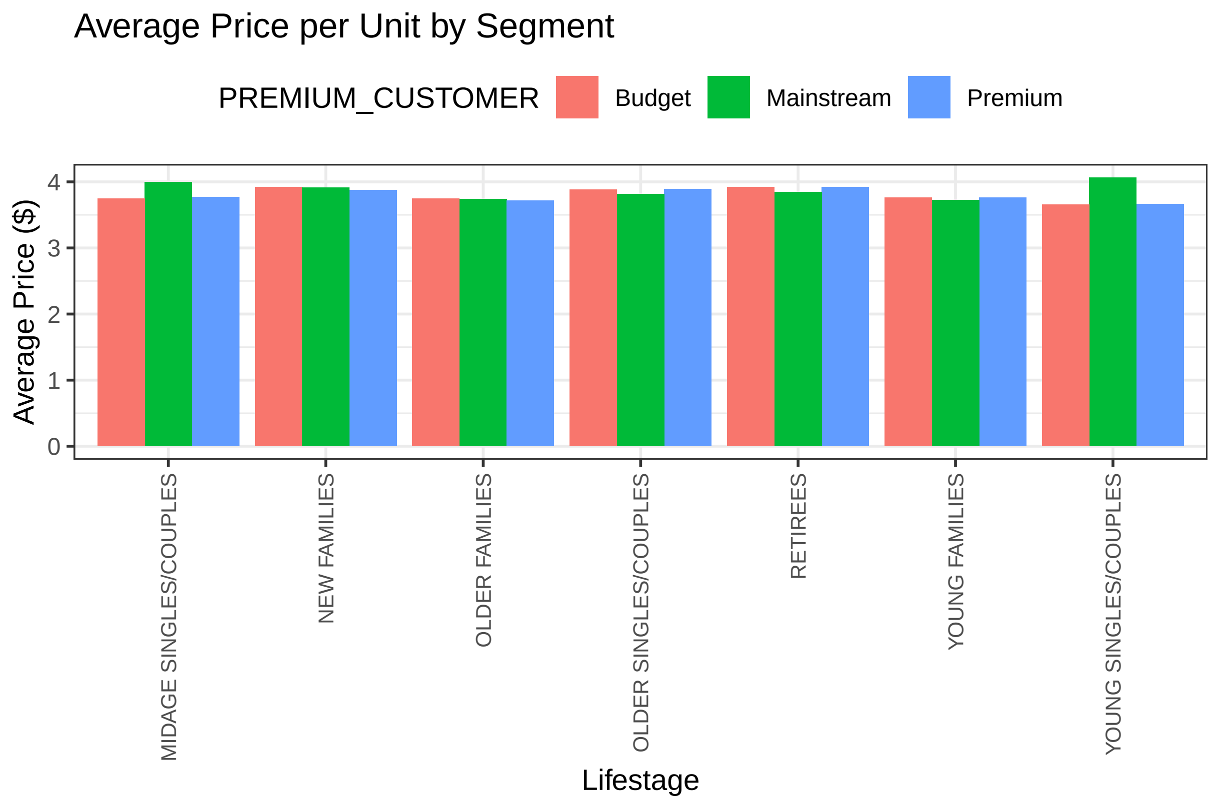
```
##
## ===== AVERAGE PRICE PER UNIT BY SEGMENT =====
```

```r
print(avg_price_print)
```

```
##                  LIFESTAGE PREMIUM_CUSTOMER UNIT_PRICE
```

```
## 14  YOUNG SINGLES/COUPLES       Mainstream       4.07
## 8  MIDAGE SINGLES/COUPLES       Mainstream       3.99
## 5                RETIREES           Budget       3.92
## 19               RETIREES          Premium       3.92
## 2            NEW FAMILIES           Budget       3.92
## 9            NEW FAMILIES       Mainstream       3.92
## 18  OLDER SINGLES/COUPLES          Premium       3.89
## 4   OLDER SINGLES/COUPLES           Budget       3.88
## 16           NEW FAMILIES          Premium       3.87
## 12               RETIREES       Mainstream       3.84
## 11  OLDER SINGLES/COUPLES       Mainstream       3.81
## 15 MIDAGE SINGLES/COUPLES          Premium       3.77
## 20         YOUNG FAMILIES          Premium       3.76
## 6          YOUNG FAMILIES           Budget       3.76
## 3          OLDER FAMILIES           Budget       3.75
## 1  MIDAGE SINGLES/COUPLES           Budget       3.74
## 10         OLDER FAMILIES       Mainstream       3.74
## 13         YOUNG FAMILIES       Mainstream       3.72
## 17         OLDER FAMILIES          Premium       3.72
## 21  YOUNG SINGLES/COUPLES          Premium       3.67
## 7   YOUNG SINGLES/COUPLES           Budget       3.66
```

```r
# Clean chart
p3 <- ggplot(avg_price, aes(x = LIFESTAGE, y = UNIT_PRICE, fill = PREMIUM_CUSTOMER)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.9)) +
  labs(title = "Average Price per Unit by Segment", y = "Average Price ($)", x = "Lifestage") +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5, size = 8),
    plot.margin = margin(b = 20, l = 10, r = 10, t = 10),
    legend.position = "top"
  )
print(p3)
```




```r
ggsave("03_avg_price_by_segment.png", p3, width = 12, height = 7, dpi = 300)
```

Deep dive: Mainstream Young Singles/Couples

```r
# Filter target segment
target <- data[LIFESTAGE == "YOUNG SINGLES/COUPLES" & PREMIUM_CUSTOMER == "Mainstream", ]

# Top 10 Brands
brand_sales <- aggregate(TOT_SALES ~ BRAND, data = target, FUN = sum)
brand_sales <- brand_sales[order(-brand_sales$TOT_SALES), ]
top_brands <- head(brand_sales, 10)

cat("\n===== TOP 10 BRANDS: MAINSTREAM YOUNG SINGLES/COUPLES =====\n")
```

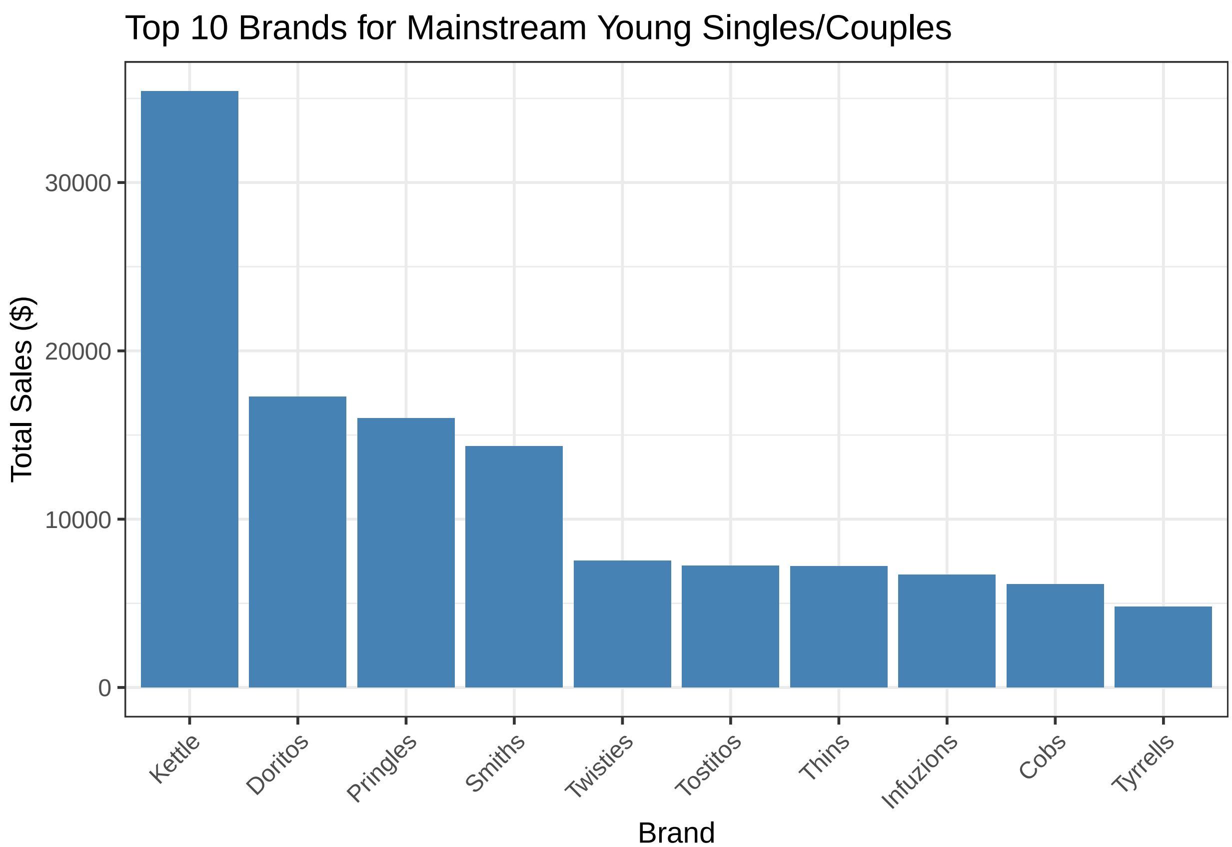
```
##
## ===== TOP 10 BRANDS: MAINSTREAM YOUNG SINGLES/COUPLES =====
```

```r
print(top_brands)
```

```
##        BRAND TOT_SALES
## 13    Kettle   35423.6
## 7    Doritos   17266.4
```

```
## 15  Pringles   16006.2
## 19    Smiths   14337.5
## 24  Twisties    7539.8
## 23  Tostitos    7238.0
## 22     Thins    7217.1
## 11 Infuzions    6693.6
## 5       Cobs    6144.6
## 25  Tyrrells    4800.6
```

```r
p4 <- ggplot(top_brands, aes(x = reorder(BRAND, -TOT_SALES), y = TOT_SALES)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Top 10 Brands for Mainstream Young Singles/Couples", x = "Brand", y = "Total Sales ($)")
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p4)
```




```r
ggsave("04_top_brands_mainstream_young.png", p4, width = 10, height = 6, dpi = 300)

# Top 5 Pack Sizes
pack_sales <- aggregate(TOT_SALES ~ PACK_SIZE, data = target, FUN = sum)
pack_sales <- pack_sales[order(-pack_sales$TOT_SALES), ]
top_packs <- head(pack_sales, 5)

cat("\n===== TOP 5 PACK SIZES: MAINSTREAM YOUNG SINGLES/COUPLES =====\n")
```

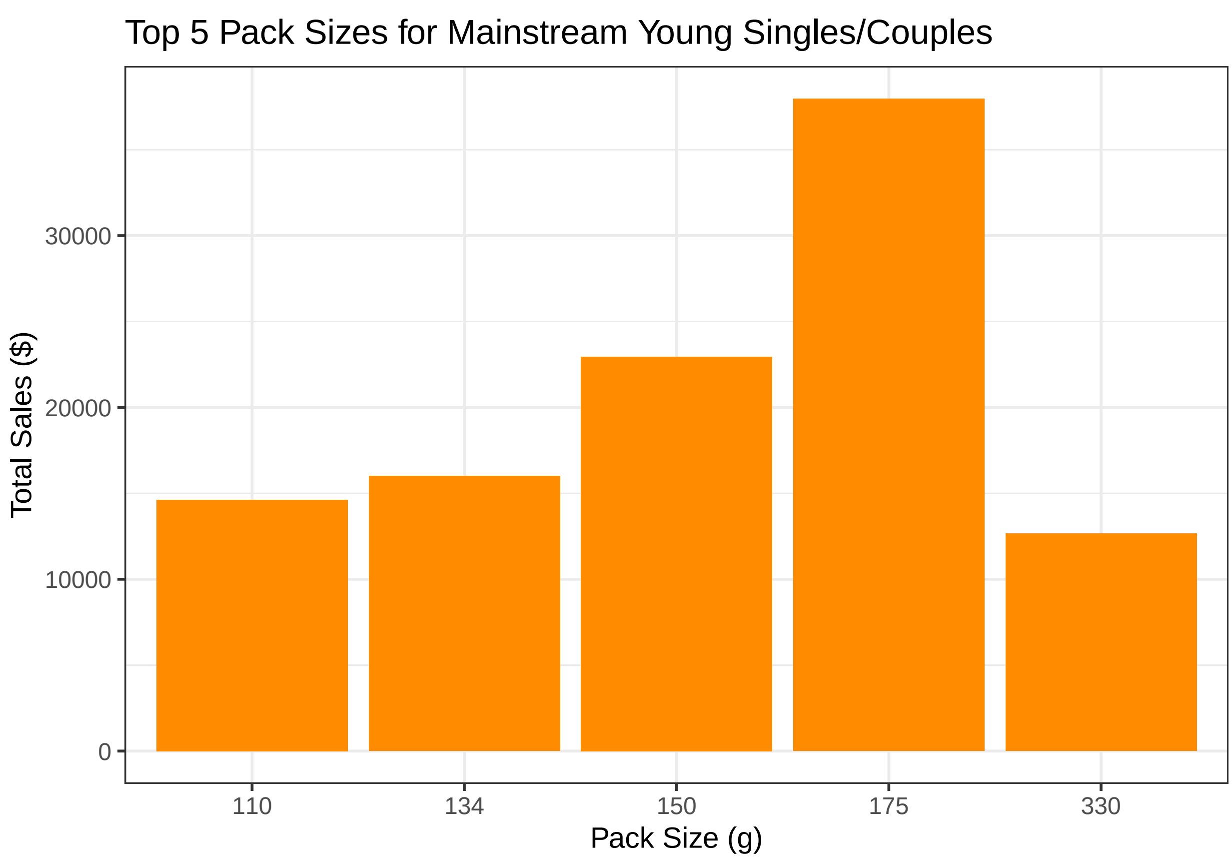
```
##
```

```
## ===== TOP 5 PACK SIZES: MAINSTREAM YOUNG SINGLES/COUPLES =====
```

```r
print(top_packs)
```

```
##    PACK_SIZE TOT_SALES
## 11       175   37967.9
## 7        150   22946.2
## 5        134   16006.2
## 3        110   14630.0
## 19       330   12654.0
```

```r
p5 <- ggplot(top_packs, aes(x = factor(PACK_SIZE), y = TOT_SALES)) +
  geom_bar(stat = "identity", fill = "darkorange") +
  labs(title = "Top 5 Pack Sizes for Mainstream Young Singles/Couples", x = "Pack Size (g)", y = "Total
  theme_bw()
print(p5)
```




```r
ggsave("05_top_pack_sizes_mainstream_young.png", p5, width = 8, height = 5, dpi = 300)
```

T-test for price difference (Mainstream vs Non-Mainstream in Singles/Couples lifestages)

```
test_data <- data[LIFESTAGE %in% c("YOUNG SINGLES/COUPLES", "MIDAGE SINGLES/COUPLES"), ]
mainstream_prices <- test_data[PREMIUM_CUSTOMER == "Mainstream", UNIT_PRICE]
non_mainstream_prices <- test_data[PREMIUM_CUSTOMER != "Mainstream", UNIT_PRICE]
t_test_result <- t.test(mainstream_prices, non_mainstream_prices)

cat("\n===== T-TEST: UNIT PRICE (MAINSTREAM VS NON-MAINSTREAM) =====\n")
```

```
##
## ===== T-TEST: UNIT PRICE (MAINSTREAM VS NON-MAINSTREAM) =====
```

```
print(t_test_result)
```

```
##
##  Welch Two Sample t-test
##
## data:  mainstream_prices and non_mainstream_prices
## t = 37.624, df = 54791, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  0.3159319 0.3506572
## sample estimates:
## mean of x mean of y
##  4.039786  3.706491
```